Baseline model evaluation

YOLOv8s fine-tuned to detect four hand gestures · Ben Nichiporik, Daffodil Ho, Joseph Lee, Thu Nguyen · November 2026

0.990
mAP50, validation (HaGRID, 160 images)

0.781
mAP50, our own photos (120 images, headline)

0.53
fist recall on our photos (weakest class, conf 0.25)

15.1 ms
per image on an RTX 5060 Ti (≈66 frames/s)

The baseline is close to perfect on held-out HaGRID images but loses 0.21 mAP50 on the photos our group collected. Most of that loss comes from **small hands in wide shots** and from **fists**, which the model either misses or confuses with a thumbs-down. The model is fast enough for a live photo booth, and a higher confidence threshold (0.55) trades a little recall for noticeably fewer false triggers.

1. What was evaluated

Model. YOLOv8s pretrained on COCO, fine-tuned on our data (transfer learning; all layers trained). 50 epochs were run; the checkpoint with the best validation fitness (epoch 41) is the baseline, `models/baseline_best.pt`.

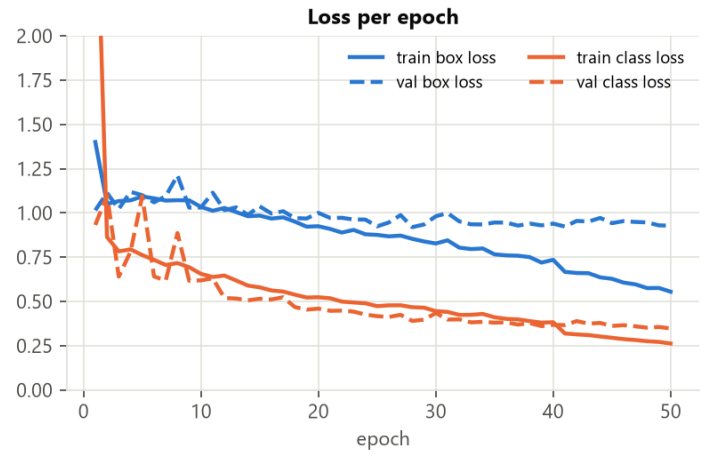
Protocol. Every model in the project is scored by the same code (`gesture_eval.py`) on three sets: HaGRID validation (used to pick the checkpoint), the full test split, and **our own photos only**, the honest estimate of photo-booth performance. Sections 4-6 then score our photos at a deployment threshold (confidence ≥ 0.25, a prediction counts as correct if its box overlaps a labelled hand with IoU ≥ 0.5 and the class matches).

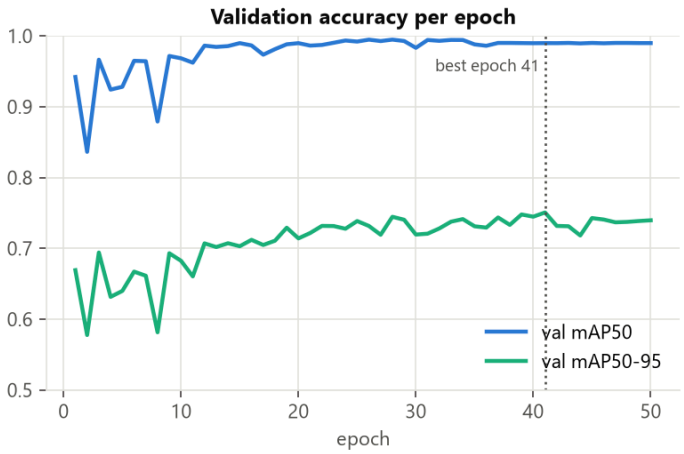
image size	640	**optimizer**	SGD
epochs	50 (patience 15)	**lr0**	0.01
batch	16	**momentum**	0.937
seed	0 (deterministic)	**weight decay**	0.0005
flips	LR 0.5, UD 0.0	**other aug.**	Ultralytics defaults (mosaic, HSV, scale)

split · source	palm	fist	like	dislike	total
train · HaGRID	320	320	320	320	1280
val · HaGRID	40	40	40	40	160
test · HaGRID	8	8	8	8	32
test · our photos	24	32	32	32	120

Images per split and class. HaGRID images 001-368 per class; our photos are 369-400 (8 per member per class).

2. Training behaviour





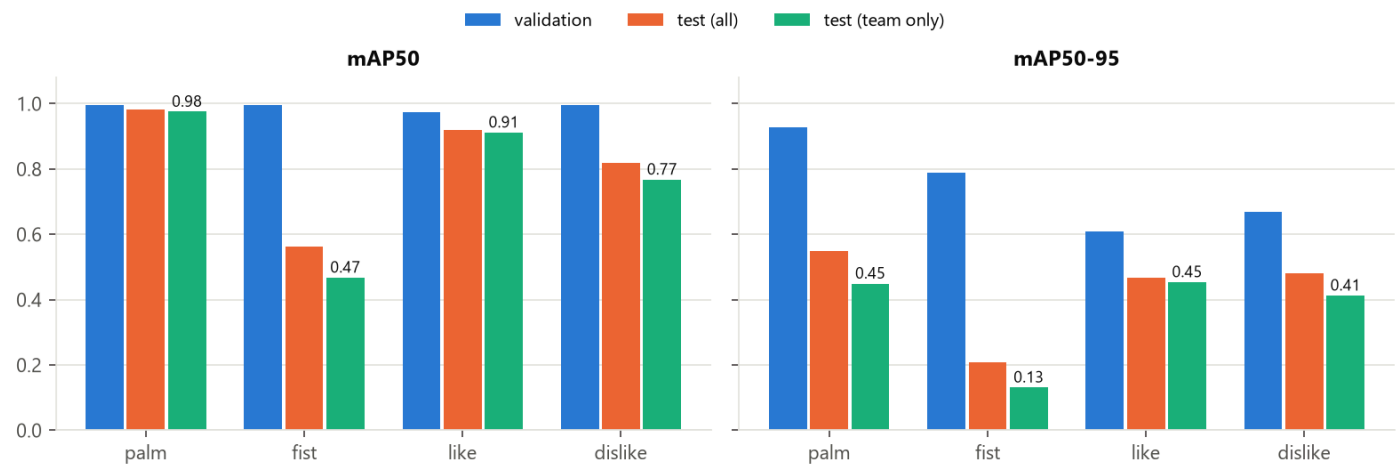
Transfer learning works quickly: validation mAP50 is already 0.94 after one epoch, because the COCO-pretrained backbone already provides general shape and edge features. Accuracy then climbs slowly. The best epoch (41) reaches mAP50 0.990 and mAP50-95 0.751. Validation losses keep falling alongside the training losses through epoch 50, so there is no sign of overfitting *to HaGRID*. Early stopping never triggered. The overfitting that does matter is to HaGRID's style of photo, which only the test on our own photos reveals.

3. Overall results

evaluation set	precision	recall	F1	mAP50	mAP50-95
validation (160)	0.990	0.984	0.987	0.990	0.748
test, all (152)	0.866	0.797	0.830	0.820	0.427
test, our photos only (120)	0.826	0.751	0.787	0.781	0.362

The validation scores are inflated: HaGRID validation images come from the same collection as the training images (one person standing and facing a phone camera, mostly indoors, one hand per photo). On our own photos, mAP50 drops by 0.21 and mAP50-95 by 0.39. The larger drop in mAP50-95 means the model usually *finds* the hand but its box doesn't line up tightly with ours. Correct detections have a mean IoU of only 0.75 with our labels. Part of this is annotation style rather than model error: each member drew boxes in MakeSense slightly differently (some include the wrist). At conf 0.25 on our photos, the 95% bootstrap intervals (resampling photos) are precision 0.71-0.85, recall 0.71-0.85 and F1 0.71-0.84. A single model's F1 is uncertain by about ±0.06, so small differences between tuned models should not be over-read.

4. Per-class performance

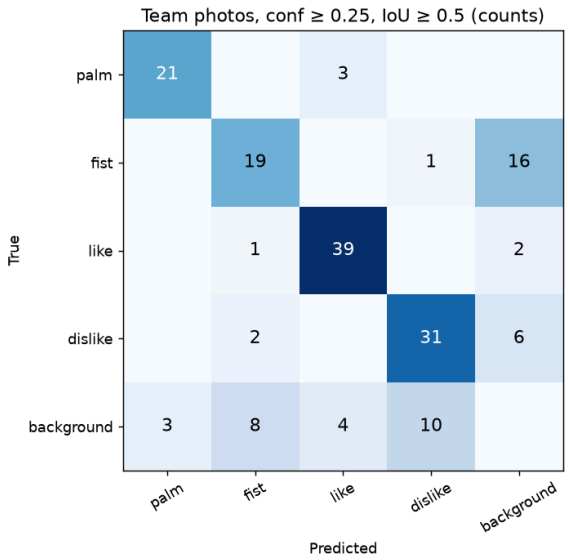


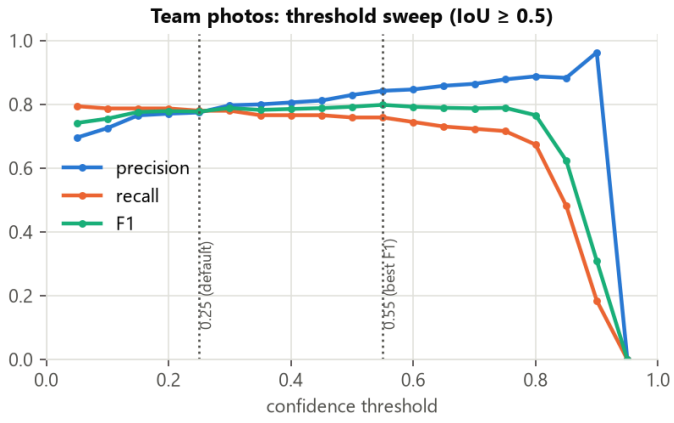
our photos	precision	recall	mAP50	mAP50-95	hands	missed @0.25	wrong class @0.25
palm	0.951	0.875	0.975	0.449	24	0	3
fist	0.653	0.528	0.468	0.131	36	16	1
like	0.877	0.833	0.912	0.454	42	2	1
dislike	0.823	0.769	0.768	0.413	39	6	2

Precision, recall and mAP from Ultralytics; missed and wrong-class counts at conf 0.25, IoU 0.5.

Palm and like transfer well: their shapes are distinctive (spread fingers, upward thumb). **Fist is the weakest class**: 16 of 36 fists are missed entirely. A fist is a compact, low-texture blob that looks like many other things at small scale, and several of our fists are raised in crowd scenes with busy backgrounds, which HaGRID never shows (Section 6). **Dislike** is confused with fist in both directions: a thumbs-down seen from the side is a fist with a short thumb, and these are the highest-confidence errors (Section 6).

5. Errors at the deployment threshold





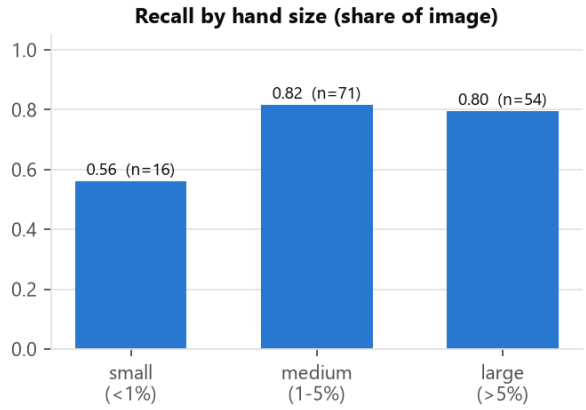
At conf 0.25 the model makes 142 predictions for 141 labelled hands in our 120 photos: **110 correct**, **24 hands missed**, **7 found with the wrong class**, and **25 predictions on background** (no labelled hand there; by predicted class: dislike 10, fist 8, like 4, palm 3). Missing a hand is the most common error. Ultralytics' own confusion matrix uses a near-zero

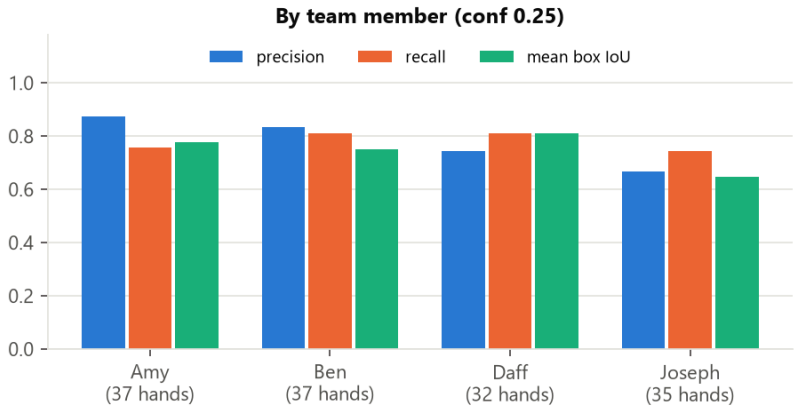
confidence threshold, so its background row looks far worse than what a user would see. The matrix here uses the real threshold.

Choosing the threshold. F1 is highest at conf 0.55 (precision 0.84, recall 0.76, F1 0.80), compared with precision 0.77 and recall 0.78 at 0.25. For a photo booth a false trigger (taking a picture nobody asked for) is worse than a missed gesture (the user just holds it a moment longer), so we recommend conf ≈ 0.55 plus requiring the gesture over several consecutive frames.

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6. Where the model fails

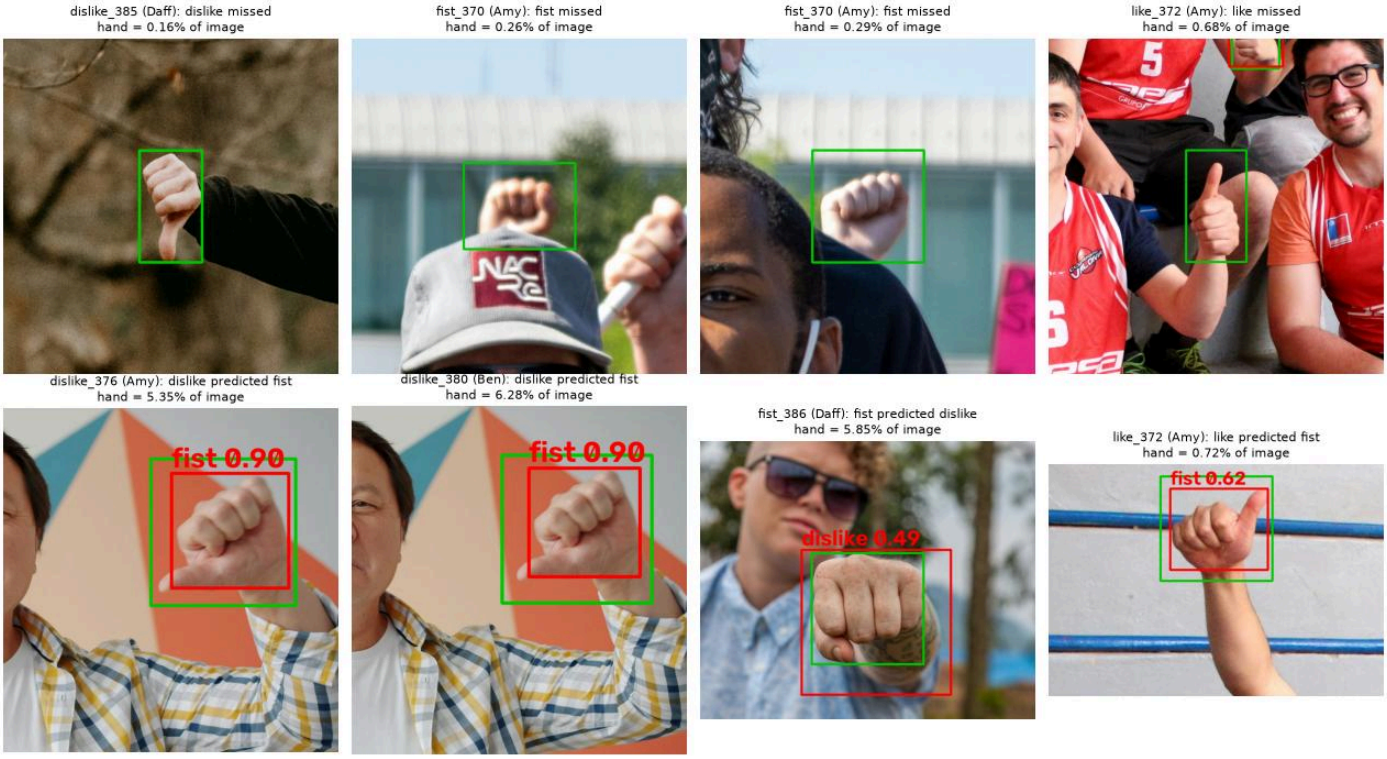




hand size	hands	recall	missed
small (<1%)	16	0.56	6
medium (1-5%)	71	0.82	12
large (>5%)	54	0.80	6

member	hands	predictions	correct	precision	recall	mean IoU
Amy	37	32	28	0.88	0.76	0.78
Ben	37	36	30	0.83	0.81	0.75
Daff	32	35	26	0.74	0.81	0.81
Joseph	35	39	26	0.67	0.74	0.65

Hand size is the clearest driver. The median labelled hand covers 3.6% of the image, but 16 hands cover under 1%, and only 56% of those are found. At imgsz 640, a hand covering 0.3% of the image is only about 35 px across after resizing, which leaves little detail. Size alone is not the cause, though: 386 of the 1280 training hands (30%) are also under 1% of the image, and the median training hand (1.6%) is smaller than ours. The difference is context. HaGRID's small hands belong to one person posing in a plain room; ours are in crowds, outdoors and at odd angles. **Per member**, the differences follow photo choice and labelling style. Amy's photos hold 10 of the 16 small hands, which lowers recall there. Joseph's photos have the lowest mean box IoU (0.65) and precision (0.67), which points to a different box style rather than a harder set of photos.



Green = our label, red = prediction (conf ≥ 0.25). Top row: smallest missed hands. Bottom row: wrong-class predictions. These are mostly fist/dislike confusions, two of them on the same duplicated photo.

7. Speed and size

Per image at imgsz 640 (batch 1, RTX 5060 Ti): preprocess 2.1 ms, inference 11.1 ms, postprocess 1.9 ms, **15.1 ms in total (≈66 frames/s)**; repeated runs vary by a few ms. The model is 23 MB (11.1 M parameters, 28.4 GFLOPs). A webcam photo booth needs about 15-30 frames/s, so a GPU has headroom to spare. CPU-only speed was not measured.

8. Limits of this evaluation

- **Small test set.** 141 hands in 120 photos. The bootstrap interval for F1 in Section 3 spans 0.13.
- **Data gap.** Amy's `palm_369` – `palm_376` have no label files yet, so the palm class has only 24 labelled hands in our photos instead of about 30.
- **Data gap.** 2 pairs of our test photos are the same picture contributed by two members (`dislike_376.jpg` (Amy) = `dislike_380.png` (Ben); `dislike_388.jpg` (Daff) = `dislike_393.jpg` (Joseph)). Those hands count twice, so the test set has 118 distinct photos, not 120.
- **Validation is optimistic.** It is HaGRID-only and drawn from the same distribution as training, so checkpoint selection does not reward robustness to our kind of photo.
- **One training seed.** Run-to-run variation was not measured; it should be kept in mind when comparing tuned models.
- **Labels differ by annotator.** Box styles vary between members, which depresses mAP50-95 and IoU-based matching.

9. Takeaways for hyperparameter tuning

1. **Small hands** are the main failure mode. We tested larger image sizes first (`Gesture_YOLO_Tuning_imgsz.ipynb`): team mAP50 was 0.769 at 480, 0.781 at 640, 0.792 at 800, 0.769 at 960, and small-hand recall did not improve. The training set is not short of small hands; it is short of our kind of scene (crowds, outdoors, several people). More varied training photos and context-changing augmentation (mosaic, copy-paste, blur) are the more promising levers.
2. **Fist vs dislike** needs more varied fist examples or more epochs of fine-grained learning; watch fist recall in every comparison.
3. Judge tuned models on **our photos** (team-only mAP50 0.781, F1 0.777 at conf 0.25), not on validation, which is saturated at 0.99.